USN. | | | | | | | | | | [24MTAD101]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY- 2025

AERODYNAMICS OF FLIGHT VEHICLES

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Consider the convergent duct and conditions with inlet and outlet velocity of 10m/s and 30m/s. If the air pressure and temperature at the inlet are P_1 1.013 bar and T_1 = 330 K, calculate the pressure at the exit.	**CO1**	**L3**	**10**
	b.	Derive the Momentum equation in the differential form	**CO1**	**L3**	**10**
2.	**a.**	Derive the fundamental equation of finite wing theory.	**CO2**	**L3**	**10**
	b.	Derive the induced angle of attack and explain how the downwash effect and induced drag are formed, including a neat sketch.	**CO3**	**L3**	**10**
3.	**a.**	Compare and contrast the characteristics of normal shocks, oblique shocks, and expansion waves in terms of their thermodynamic effects on flow properties. Provide examples of where each phenomenon occurs in practical applications.	**CO4**	**L2**	**10**
	b.	Air expands through CD Nozzle to 100kPa from a large reservoir with pressure and temperature of 600kPa and 40°C respectively. Find **i.** Ratio of nozzle exit area. **ii.** Discharge velocity under design consideration. **iii.** At what back pressure will there be a normal shock at the exit plane of the nozzle.	**CO4**	**L3**	**10**
		OR			
4.	**a.**	Derive the Prandtl-Meyer Relation for the expansion fan.	**CO4**	**L3**	**10**
	b.	The flat plate kept at 15° angle of attack to a supersonic flow stream at a Mach number of 2.4. Solve the flow fluid around the plate.	**CO4**	**L3**	**10**
5.	**a.**	A supersonic wind tunnel nozzle is to be designed for M = 2.5 with a test-section 1m^2 in area. The supply pressure and temperature at the nozzle inlet are 700kN/m^2 and 27°C respectively where the velocity is	**CO5**	**L3**	**10**

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		negligible. Assume that the flow is isentropic with $\gamma = 1.4$. Also consider one-dimensional at throat-and test-section. Calculate : **i.** Throat Area **ii.** Temperature at throat **iii.** Velocity of flow at test section **iv.** Mass flow rate through the test section			
	b.	Derive the Area-Mach relation for the Delaval Nozzle.	**CO5**	**L3**	**10**
		OR			
6.	**a.**	Derive the relation for the Prandtl-Glauert Rule and provide the relation for the indirect Problem of Transformation.	**CO5**	**L3**	**10**
	b.	Derive the Von Karman Rule for Transonic flow	**CO5**	**L3**	**10**
7.	**a.**	How does the Reynolds number influence the development and characteristics of a boundary layer?	**CO6**	**L2**	**10**
	b.	What are the key features of turbulent flow, and how is it distinct from laminar flow?	**CO6**	**L2**	**10**
		OR			
8.	**a.**	Explain the layers of a turbulent boundary layer and the roles of the viscous sub-layer and buffer layer.	**CO6**	**L2**	**10**
	b.	State the logarithmic velocity profile equation for turbulent boundary layers and discuss its importance.	**CO6**	**L2**	**10**